

Supplementary Material

Case implementation and provenance for the six-gate audit

This supplement accompanies “A Benchmark-Integrity Audit Protocol for Conformal Battery State-of-Health Evaluation: Recent-Label Dependence, Schema-Bundle Sensitivity, and Threshold-Decision Risk.” It makes the retrospective case implementation auditable. The integrated G1–G6 synthesis was developed after the diagnostic program; it was not prospectively registered. The component plan files, scripts, and current paper data are preserved at immutable public replication commit 42f1120ed1f5cbc0282a706ebd4bec24a8d19d6e.

Table S1. Analysis-to-gate implementation map

ID	Governed analysis	Gate(s)	Tasks and rows	Feature schema	Model / conformal family
A01 / exp001	Pooled-domain, cell-held-out coverage baseline	G1, G2, G3	Main; governed cell-disjoint train/calibration/validation/test partitions	QA numeric schema; recent prev_soh permitted only as a declared sequential measurement	Ridge, gradient boosting (GB), random forest, and extra trees with standard, weighted, domain-Mondrian, online-adaptive, and persistence-anchor conformal candidates; selected diagnostic is persistence-anchor protocol-Mondrian
A02 / exp002	Recent-label and resistance ablation	G1	Main; identical governed partitions across four conditions	Full; minus prev_soh; minus resistance; minus both; persistence candidate disabled when prev_soh is absent	Same candidate families as A01; validation-only selection within each condition
A03 / exp003	Cross-dataset transfer diagnostic	G1, G2, G3	LCAL, TCAL, LNAS, TNAS, LOxf, TOxf	Cross-domain numeric schema excludes dataset-, chemistry-, and protocol-identity columns; recent label remains a declared condition	Same point-predictor and conformal candidate families as A01, including protocol-Mondrian where groups exist
A04 / exp004	Residual-bias stress screen	Ancillary to G3/G4 interpretation	Main test rows from A01	No refit; additive bias is applied to the selected point prediction	Selected A01 persistence-anchor interval; conformal radius is fixed
A05 / exp005	Host-timing screen	Ancillary only	512 real Main test rows	prev_soh and precomputed calibration-group radius	Persistence-only lookup, persistence-anchor interval lookup, and GB inference; no deployment claim
A06 / exp006	Method-family and dependence-aware comparison	G2, G3	Main plus all six cross-dataset tasks	Governed task-specific schemas from the split manifest	Persistence-anchor protocol-Mondrian, GB quantile regression (QR), GB conformalized quantile regression (CQR), and NGBoost-CQR
A07 / exp007	Cross-family UQ and two-threshold decision audit	G2, G3, G5	Same seven tasks and corrected disjoint rows	Same task schemas as A06	Persistence-anchor, GB-QR, GB-CQR, and NGBoost-CQR; decision cells at SOH 0.70 and 0.80
A08 / exp008	Reliability and decision-consequence audit	G1, G2, G3, G5	Same seven tasks and corrected disjoint rows	Declared recent-label availability and task schemas	A07 families; cell-block uncertainty, subgroup checks, false acceptance, false rejection, and uncertainty
A09 / exp009	Schema-cleaning and residual-shift audit	G4, G5	Same seven tasks; added harmonized LNAS diagnostic	Original task schema versus a common cross-dataset schema	GB-CQR and supporting A08 families; identical rows within contrasts
A10 / exp010	No-recent-label hard-regime audit	G1, G4, G5	Seven tasks; horizons 0, 5, and 20; six SOH thresholds	No prev_soh; original, common-cycle, and stage schemas	GB-QR, GB-CQR, and stage-Mondrian CQR
A11 / exp011	Schema-to-decision audit and independent control	G1, G4, G5	Seven horizon-0 tasks; identical within-task rows; 245 contrast/end-point tests	As-used bundle, clean-original, common-cycle no-recent, common-cycle plus recent-label reference, and clean plus a fixed-seed permuted log-cycle control	One GB-CQR procedure fitted under each schema; paired cell-block deltas
A12 / exp012	Localized-residual sensitivity comparator	G6	Same seven corrected tasks and identical standard/localized rows	Clean task features shared by calibration and target rows; no density-ratio estimate	Standard GB-CQR versus Gaussian-kernel localized-residual CQR; sensitivity diagnostic only

Table S2. Locked parameters, selection, plan provenance, rerun, and deviations

ID	Hyperparameters / search grid	Calibration and selection rule	Plan identifier, archive date, immutable plan commit	Governed run identifier	Recorded deviation or claim boundary
A01	Seed 42; GB: 250 trees, depth 3, learning rate 0.035; RF/ET: 300 trees, minimum leaf 3; q grid 0.90, 0.93, 0.95, 0.97, 0.99, 1.00	Validation coverage first, then minimum validation width; test metrics excluded	experiments/exp001_main/plan.md; 2026-05-20; e2ed4beba44f49f92f36f3c66a545bdf528eb010	PCR-2026-07-18-01 full real-data rerun; scripts/run_real_experiments.py --seed 42	Legacy experiment lacks a standalone run manifest; current manuscript values were regenerated in the pinned full rerun and are covered by the paper SSOT/raw recomputation gate

ID	Hyperparameters / search grid	Calibration and selection rule	Plan identifier, archive date, immutable plan commit	Governed run identifier	Recorded deviation or claim boundary
A02	Seed 42; four predeclared drop-feature conditions; A01 model and q grids	A01 validation-only rule, applied separately within each condition	experiments/exp002_ablation/plan.md; 2026-05-20; e2ed4beba44f9f2f36f3c66a545bdf528eb010	PCR-2026-07-18-01 full real-data rerun	The selected model family can change across conditions; therefore the Main contrast is pipeline sensitivity, not a method-matched feature effect
A03	Seed 42; A01 model/q grids; target windows 0–20% train, 20–35% calibration, 35–55% validation, >55% test	Online candidate preferred only if validation coverage is at least 0.90; otherwise coverage-first/width selection	experiments/exp003_cross_protocol/plan.md; 2026-05-20; e2ed4beba44f9f2f36f3c66a545bdf528eb010	PCR-2026-07-18-01 full real-data rerun	Negative transfer results were retained; no broad-transfer success claim
A04	Bias grid –0.05, –0.02, –0.01, –0.005, 0, +0.005, +0.01, +0.02, +0.05	No tuning or radius refit after bias injection	experiments/exp004_stress_failure/plan.md; 2026-05-20; e2ed4beba44f9f2f36f3c66a545bdf528eb010	PCR-2026-07-18-01 full real-data rerun	Prediction-bias robustness screen only; not evidence of a physical degradation mechanism
A05	20 warmups, 300 timed repetitions, 512 real rows	Fixed timed path; no model selection	experiments/exp005_edge/plan.md; 2026-05-20; e2ed4beba44f9f2f36f3c66a545bdf528eb010	PCR-2026-07-18-01 full real-data rerun	Host amortized throughput only; explicitly excluded from edge/deployment evidence
A06	Seed 42; 1,000 cell-block resamples; q grid 0.90–1.00; GB quantiles 0.05/0.95 with 250 trees, depth 3, learning rate 0.035; NGBoost 300 estimators, learning rate 0.03	Validation coverage first, then width; contribution positioning fixed before result interpretation	experiments/exp006_fpa_repair/plan.md; 2026-05-30; 83d81d8c59c893979c57ee55b62912e0051ddc23	experiments/exp006_fpa_repair/results/run_manifest.json; corrected rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe15	Closed as a diagnostic comparison rather than method superiority
A07	Seed 42; 1,000 block resamples; q/GB/NGBoost settings as A06; denominator floor 5	Validation coverage-first width; inference intervals do not tune q	experiments/exp007_fpa_round2_repair/plan.md; 2026-05-30; 83d81d8c59c893979c57ee55b62912e0051ddc23	A07 run manifest; corrected rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe15	July repair made target train/calibration/validation/test windows pairwise disjoint and superseded earlier overlapping outputs
A08	Seed 42; 1,000 block resamples; denominator floor 5; A07 method settings	Same-row method comparison; failed cells/tasks retained	experiments/exp008_reliability_audit/plan.md; 2026-06-01; ba5080bc38f9717aca5919651b22db141005f95b	A08 run manifest; corrected rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe15	July repair preserves cluster multiplicity in cell resampling; thresholds remain illustrative decision cutoffs
A09	Seed 42; 1,000 block resamples; denominator floor 5; q grid 0.90–1.00	Identical rows and coverage-aligned comparisons; common-schema diagnostic added for LNAS	experiments/exp009_fpa_round4_repair/plan.md; 2026-06-01; ba5080bc38f9717aca5919651b22db141005f95b	A09 run manifest; corrected rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe15	Common-schema cleaning cannot eliminate residual protocol drift; no physical-mechanism attribution
A10	Seed 42; 1,000 block resamples; horizons 0/5/20; thresholds 0.65/0.70/0.75/0.80/0.85/0.90; denominator floor 5; q grid 0.90–1.00	Validation-only q selection; Bonferroni and BH multiplicity summaries	experiments/exp010_hard_regime_audit/plan.md; 2026-06-01; ba5080bc38f9717aca5919651b22db141005f95b	A10 run manifest; corrected rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe15	As-used no-recent schema rows containing bookkeeping fields are treated as contaminated and superseded by A11 contrasts
A11	Seed 42; control-column seed 1033; 1,000 paired cell-block resamples; family size 5×7×7=245; family-wise alpha 0.05; denominator floor 5; GB settings as A06	Identical task rows; Bonferroni-adjusted normal intervals from block-bootstrap SE; exact paired-cell sign test at nonzero zero-SE boundaries	experiments/exp011_original_paper_substance/plan.md; 2026-06-01; ba5080bc38f9717aca5919651b22db141005f95b	A11 run manifest; corrected rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe15	The July repair replaced the earlier duplicated-fit control with an independently permuted log-cycle control. Because that single-draw control is non-null in LCAL/TCAL, it is a veto only and blocks mechanism attribution
A12	Seed 42; 1,000 block resamples; denominator floor 5; bandwidth grid 0.25, 0.50, 1, 2, 4, infinity	Validation-only coverage-first width selection; test metrics excluded	experiments/exp012_shift_adaptive_cp_comparator/plan.md; 2026-06-01; ba5080bc38f9717aca5919651b22db141005f95b	A12 run manifest; corrected rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe15	The implemented comparator is localized residual weighting, not a density-ratio or guarantee-bearing shift-adaptive conformal method; one point-coverage failure recovers, no lower-bound shortfall recovers

Provenance boundary

The plan-archive commits above precede the governed July 2026 rerun that supplies the current manuscript values. They establish that component analysis rules existed before the corrected rerun; they do not make the integrated six-gate synthesis prospectively registered. The earlier exploratory processed CSV could not be recovered byte-for-byte. The paper therefore reports only the corrected rerun regenerated from preserved raw holdings under Python 3.11.15, NumPy 2.2.6, pandas 2.3.3, and scikit-learn 1.7.2. Supplementary claims must be interpreted with the same case-bounded scope as the manuscript.